CBCS SCHEME

USN ☐☐☐☐☐☐☐☐☐☐ **BESCK104D**

First Semester B.E./B.Tech. Degree Examination, Dec.2024/Jan.2025

Introduction to Mechanical Engineering

Time: 3 hrs. Max. Marks: 100

Note: 1. Answer any FIVE full questions, choosing ONE full question from each module.
2. M : Marks , L: Bloom's level , C: Course outcomes.

		Module – 1	**M**	**L**	**C**
Q.1	**a.**	Explain the role of Mechanical Engineering in Industries and society.	**10**	**L2**	**CO1**
	b.	Explain briefly the emerging trends of Mechanical Engineering in manufacturing and Energy sector.	**10**	**L2**	**CO1**
		OR			
Q.2	**a.**	With sketch, explain the working principal of Thermal Power Plant.	**10**	**L2**	**CO1**
	b.	Explain the utilization of solar energy using Flat plate collector with a schematic diagram.	**10**	**L2**	**CO1**
		Module – 2			
Q.3	**a.**	With line diagram, explain the working principle of Lathe machine.	**6**	**L2**	**CO2**
	b.	With neat sketches, illustrate i) Boring operation and ii) Reaming operation of drilling machine.	**8**	**L3**	**CO2**
	c.	Differentiate between Up milling and Down milling.	**6**	**L2**	**CO2**
		OR			
Q.4	**a.**	With sketch, explain the basic components of CNC machine.	**10**	**L2**	**CO2**
	b.	State the advantages and applications of CNC machine.	**5**	**L1**	**CO2**
	c.	List and explain in brief the various steps involved in Additive manufacturing.	**5**	**L1**	**CO2**
		Module – 3			
Q.5	**a.**	Analyze the working of four stroke petrol engine with sketches. Plot the PV diagram.	**10**	**L4**	**CO3**
	b.	List any five significant difference between petrol and diesel engine.	**5**	**L1**	**CO3**
	c.	State the various applications of IC engines.	**5**	**L1**	**CO3**
		OR			
Q.6	**a.**	Describe Electrical Vehicles (EVs). Explain the components and working of Electrical vehicles.	**10**	**L2**	**CO3**

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	b.	List the advantages and disadvantages of Electrical and Hybrid Electrical Vehicles.	10	L1	CO3
		Module – 4			
Q.7	a.	State the classification of Metals.	5	L1	CO4
	b.	Discuss the compositions , properties and applications of Ferrous and Non – Ferrous metals.	10	L2	CO4
	c.	What are Shape Memory Alloys? Mention the applications of Shape Memory alloys.	5	L2	CO4
		OR			
Q.8	a.	What is Welding? With sketch, explain the process of Arc welding.	10	L2	CO4
	b.	With figures, explain the 3 – types of flames in Oxy – Acetylene gas welding.	6	L1	CO4
	c.	Discuss the application of brazing.	4	L2	CO4
		Module – 5			
Q.9	a.	Define Automation. Explain 3 – types of automation with example.	10	L2	CO5
	b.	With sketches, explain Cartesian and Polar robotic configurations.	6	L2	CO5
	c.	Differentiate between Open loop and Closed loop control system.	4	L2	CO5
		OR			
Q.10	a.	Define Internet of Things (IoT). Briefly explain the characteristics of IoT.	10	L2	CO5
	b.	Explain physical design and protocols of IoT.	10	L2	CO5

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